

Title	Revolutionizing Organic Waste Management Through IoT-Integrated Smart Composting: A Sustainable Solution for Waste Reduction and Soil Enrichment	
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CONTEXT When does the problem occur? - The problem happens when household, community, and small business generate large amount of organic waste that affects communities because of the improper disposal of biodegradable waste and it leads to increased landfill use.	PROBLEM What is the root cause of the problem? - The root cause of the problem is that most of the households and even the community don't know the proper methods to manage their organic waste. And the result, people often dispose of their biodegradable waste incorrectly.	ALTERNATIVES What do customers do now to fix the problem? - Households often attempt backyard composting or resort to selling recyclables, though results are inconsistent. The City ENRO also implements vermiculture and operates a large composting machine, but these face operational and power-related challenges.
CUSTOMERS Who has the problem most often? - The City ENRO, which is responsible for managing waste across all barangays, is most heavily affected by the issue since they carry the biggest responsibility in disposal and processing.	EMOTIONAL IMPACT How does the customer feel? - They City ENRO feels frustrated and pressured because of the responsibility of managing the waste of community. The users will feel satisfaction because they no longer have to worry about where to dispose their organic waste, as the SCB makes it much easier to process and break the organic waste. QUANTIFIABLE IMPACT What is the measurable impact (include units)? - According to City ENRO about 55-56% of the city's waste is biodegradable, which shows how much of the total waste is made up of organic waste that ends up in landfills and contribute to environmental pollution.	ALTERNATIVE SHORTCOMINGS What are the disadvantages of the alternatives? - Backyard composting is difficult to maintain because of limited space, lack of knowledge, and little monitoring, which often leads to poor results. The city's composting machine cannot be relied upon since power issues frequently prevent it from working properly. Although vermiculture can be effective, it is too small in scale to handle the large amount of waste produced by communities.

PROBLEM STATEMENT CANVAS

